

## FLAGSHIP SOLUTION SCALING TRACK PROFILE

# Closing the Adaptation Gap: Scaling Agvisely Climate Services for Resilient Farming in Bangladesh

Scaling for Impact Science Program | Area of Work 2

**Grand Challenge 1** · Close the Climate Adaptation Gap through Scalable Climate Services and Smart Water Management

**Flagship 1.1** · Digital Climate Services for Resilient Farm & Enterprise Management

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# Foreword

## CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

**Flagship 1.1** — Digital Climate Services for Resilient Farm & Enterprise Management

**Flagship 1.2** — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

**Flagship 2.1** — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

**Flagship 2.2** — Entrepreneurship through Mechanization & Postharvest Services

**Flagship 2.3** — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

**Flagship 3.1** — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

# 1. Overview

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Smallholder farmers in Bangladesh face intensifying climate threats—erratic rainfall, temperature extremes, cyclones, floods, droughts, and salinity—that disrupt crops, livestock, and aquaculture. Existing advisory systems are fragmented, manual, poorly localized, and insufficiently inclusive, leaving most farmers without timely, actionable guidance. This scaling track aims to close this adaptation gap by scaling Agvisely, an automated climate service that delivers hyper-localized forecasts and farm-specific advisories to five million farmers by 2028, with a focus on inclusion for women, youth, and marginalized groups. The primary geographic focus is all 495 sub-districts (Upazilas) of Bangladesh.

The core solution is a bundled innovation package centered on the Agvisely platform (). Agvisely automates the translation of weather forecast data into actionable advisories by linking it to biological threshold models for crops, livestock, and aquaculture. It delivers stage-specific good agricultural practices and integrated pest management advice directly to extension officers and, increasingly, to farmers via multiple channels. This represents a systemic shift from manual, committee-driven advisories to an automated, data-driven, and scalable system.

The evidence base for scaling is robust. Agvisely is already endorsed by the Ministry of Agriculture (2019) and mandated for use by the Department of Agricultural Extension (DAE). In latest estimate it reaches 1,446,174 farmers through 8,368 registered extension officers. Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) data confirms that farmers using Agvisely adapt their management practices more effectively to weather risks. This scaling track is a collaboration between the International Maize and Wheat Improvement Center (CIMMYT, lead), the International Rice Research Institute (IRRI), WorldFish, and key government, non-government and private sector partners.

## 2. Innovation Bundle and Scaling Ambition

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**What is Being Scaled** The core innovation being scaled is the **Agvisely bundled advisory service**, which includes:

- **Automated Climate Advisories:** Hyper-localized, 5-day weather forecasts and risk alerts for crops (rice, wheat, maize, lentils, mungbean, mustard, potato), livestock (cattle, poultry), and aquaculture (Rohu, Tilapia, carp, shrimp/prawn).
- **Good Agricultural/Livestock/Aquaculture Practices (GAPs):** Stage-specific management recommendations for enhancing resilience and productivity.
- **Integrated Pest Management (IPM) Advisories:** Pest and disease risk warnings and management suggestions.
- **API Infrastructure/ Decision tree integration:** The technical backbone enabling integration with public and private digital platforms (e.g., BAMIS, Khamari, OneFish, iFarmer) to broaden reach.

### Evidence of Impact

- **Reach:** Currently reaches 1,446,174 farmers through 8,368 registered extension officers, with a mandate under consideration from the Department of Agricultural Extension (DAE) for its officers to use the app.
- **Effectiveness:** MELIA data demonstrates that farmers using Agvisely advisories adapt their management practices more effectively to weather risks compared to non-users.
- **Institutional Adoption:** The Department of Agricultural Extension (DAE) has issued a circular mandating its extension officers to use Agvisely, demonstrating strong government buy-in and a pathway to institutionalization.

## Scaling Targets

| Outcome Metric                   |           | 2026 Target                     | Long-term Goal (3–5 years)         |
|----------------------------------|-----------|---------------------------------|------------------------------------|
| Farmers reached                  |           | 1 million (cumulative)          | 3.5 million (30% active users)     |
| Women farmers reached            |           | 35% of total (350,000)          | 35% of total (1.75 million)        |
| Extension staff/partners trained |           | 4,000                           | N/A (embedded in systems)          |
| Public integrated                | platforms | 2 (BAMIS, Khamari)              | 5 (incl. AIS call center, OneFish) |
| Private sector integrations      | API       | 2 (e.g., iFarmer, Genius Farms) | 5+ (incl. finance and insurance)   |

## 3. Scaling Pathway and Theory of Change

**Pathway Strategy** The scaling strategy employs a multi-pronged pathway focused on "scaling out" and "scaling deep."

- **Institutionalization (Scaling Deep):** The primary pathway is embedding Agvisely within the core workflows of public sector agencies like DAE, Department of Livestock Services (DLS), and Department of Fisheries (DoF). This is achieved through formal agreements, API integration or decision tree integration into their official platforms (e.g., BAMIS, Khamari app), and training of extension officers to ensure long-term sustainability and ownership.
- **Market-Based Scaling (Scaling Out):** Alongside public integration, the track will leverage private sector and NGO partners to expand reach. This involves making Agvisely's API available to fintech companies (e.g., iFarmer, PKSF), insurtech providers (e.g., Green Delta), agribusinesses, and NGOs. By bundling advisories with financial products, insurance, and mechanization services, the value proposition is enhanced for both partners and end-users.
- **Inclusive Access:** To overcome digital literacy barriers, the pathway includes developing an AI-driven text-to-voice service in Bangla and regional dialects, making advisories accessible via advisory hotlines for low-literacy farmers.

**Theory of Change** If smallholders gain access to bundled, localized climate advisories through public, private (including service providers), and NGO extension channels, **and** key barriers such as digital literacy (via AI voice services) and last-mile input access (via bundled services) are addressed, **then** farmers will adopt climate-adaptive practices, reducing production risks and losses, **leading to** increased productivity, income, and resilience for farming households, **ultimately contributing to** a national farming system that is systematically better prepared for climate change.

### Key Activities Funded by S4I

- **Deepening Institutional Integration:** Formalizing API or decision tree integration with DAE's BAMIS platform, BARC's Khamari app, and the AIS call center/SMS system to automate and replace manual advisory processes. Integrating wheat and maize disease early warning and advisories to the Agvisely platform to enhance the reach of the farmers.

- **Catalyzing Private Sector Bundles:** Facilitating and supporting API integration or use of agvisely app/ AI hotline with key private sector partners in finance (eg: PKSF, iFarmer) and insurance (eg: Green Delta, Adorsho Pranisheba Ltd.) to pilot and scale bundled advisory products.
- **Enabling Inclusive Access:** Prototyping and deploying an AI-enabled text-to-voice service in Bangla and regional dialects to be hosted by the Agricultural Information Service (AIS), ensuring access for low-literacy farmers. Possible scaling path is the institutionalization with Agricultural Information Service.
- **Strengthening NGO Partnerships:** Formalizing agreements and training NGO partners (e.g., BRAC, CNRS, ASA) to integrate Agvisely into their existing programs.
- **Strategic Capacity Building:** Training 4,000 extension staff, lead farmers, and private sector actors to effectively use and disseminate Agvisely advisories, reinforcing institutional ownership.

## 4. Partners and Roles

| Partner Type            | Specific Partners and Quantities                                                                                                                                                                                              | Role in Scaling Pathway                                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Demand Partners</b>  | <b>Private Sector:</b> iFarmer (0.26M farmers), Genius Farms Ltd., Green Delta Insurance, Adorsho Pranisheba Ltd., InsureCow Ltd. <b>Finance:</b> PKSF (apex body for microfinance). <b>NGOs:</b> BRAC, CNRS, ASA, Shushilan. | Integrate Agvisely API or scale the use of the Agvisley mobile app/ the AI voice platform into their platforms to offer bundled services (loans, insurance, pest detection) to their clients, creating demand and expanding reach. Co-invest in delivery. |
| <b>Scaling Partners</b> | <b>Government:</b> DAE (14k+ officers), DLS, DoF, AIS, BARC, BMD, BLRI, RIMES BFRI. <b>Research:</b> CIMMYT (lead), IRRI, WorldFish. <b>Multilateral:</b> World Bank (PARTNER project), IFAD, ADB.                            | Adopt and institutionalize Agvisely or its' decision trees in public extension and advisory systems. Co-develop content, validate thresholds, and provide meteorological data. Provide programmatic platforms for large-scale farmer reach.               |

**Funding Model:** S4I catalytic funding focuses on enabling system-level changes—fostering partnerships, supporting API integrations or institutionalizing the use of agvisely, and developing inclusive AI tools. The core costs of service delivery (e.g., extension officer time, private sector platforms, government infrastructure) are financed by the respective partners. Recurring costs for servers and updates are intended to be absorbed by government and private sector partners post-institutionalization, with cost-recovery models explored for private API access.

## 5. Enabling Environment and Risks

### Opportunities

- **Strong Government Mandate:** A circular from the Ministry of Agriculture and DAE already mandates the use of Agvisely, providing a powerful policy lever for institutionalization.
- **High Partner Demand:** Formal letters of demand and interest from major private sector (iFarmer, Genius Farms) and financial (PKSF, Green Delta) actors demonstrate a clear market pull.



- **Alignment with Major Programs:** Agvisely is strategically positioned to be integrated into large-scale multilateral investments, including the World Bank's PARTNER/Agri-connect programs and potential IFAD/ADB programs.
- **Proven Technical Model:** The automated threshold-based system is scientifically validated and ready for integration with multiple platforms via its API or through conversational AI systems, once validated.

## Challenges and Mitigation

- **Institutional Risk:** Shifting priorities or leadership changes in government agencies could derail institutionalization.
- **Mitigation:** Embed Agvisely in formal annual workplans and budgets, not just mandates. Continuously train a broad base of staff to ensure ownership transcends individual leaders. Integrating the agvisely decision trees for the existing Government apps like BAMIS.
- **Financial Sustainability:** Long-term reliance on donor funding for core operations.
- **Mitigation:** Transfer maintenance costs to government partners post-integration of decision trees. Develop a tiered API access model, offering free basic access to public/non-profits and fee-based premium access for commercial entities.
- **Operational Barrier:** Low digital literacy among farmers limits direct app usage.
- **Mitigation:** Prioritize the development of the AI text-to-voice service to make advisories accessible via basic phones and hotlines.
- **Technical barriers:** Use of aquaculture advisories need reference pond temperatures uploaded to the system.

**Mitigation:** Use of sensors (IoT) to create dynamic data flow of water temperature and other parameters to Agvisely system.

- **Social Barrier:** Gender gaps in technology access could exclude women.
- **Mitigation:** Set an explicit target of 35% women users and promote women-to-women service provider models through NGO partners to enhance trust and accessibility.

## 6. Learning and Adaptation

**Monitoring Focus** The MELIA framework will track progress and generate insights for adaptive management, focusing on:

- **Process Outcomes:** Are API/ decision tree integrations on track? Are partnership agreements formalized? Are training targets being met?
- **Behavior Change:** Are extension officers and private partners actively using and disseminating Agvisely advisories? Are farmers taking new actions based on the advice?
- **System Outcomes:** Is Agvisely being embedded into government workflows? Are private partners developing new, viable bundled products? What is the verified reach to women and marginalized groups?

**Adaptive Management** Learning will be translated into action through:

- **Quarterly Progress Reviews** with the S4I AoW2 team to assess progress against milestones and troubleshoot bottlenecks.
- **Partner Feedback Loops:** Regular check-ins with DAE, private sector partners, and NGOs to understand their challenges and adapt the integration or training support accordingly.

- **MELIA Study:** A dedicated study will be conducted to rigorously assess user adoption, implementation constraints, and early impact, feeding findings directly into the scaling strategy.

## 7. Next Steps and Timeline (2026)

| Quarter | Key Activities                                                                                                                                                                                                                                                                                                      |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1-Q3   | Formalize agreements with DAE, DLS, AIS, and BARC for API integration/ decision tree integration and co-branded delivery. Initiate API integration with private sector partners (iFarmer, Genius Farms).                                                                                                            |
| Q2-Q3   | Complete integration of Agvisely API / decision trees into BAMIS or Khamari applications. Secure agreements with One Fish App and initial NGO partners (e.g., BRAC). Begin development of AI text-to-voice prototype. Training of machinery service providers for climate smart planting and harvesting operations. |
| Q3-Q4   | Promoting use of Agvisely / AI hotline with AIS call center. Extend API access/ training of Agvisley app to additional non-profit and for-profit service providers. Pilot integration of advisories with financial/insurance products (e.g., with PKSF, Green Delta).                                               |
| Q4      | Launch farmer enrollment and outreach campaigns via extension networks and agribusiness partners. Begin training of 4,000 extension staff, lead farmers, and service providers. Complete AI voice prototype testing.                                                                                                |

**Communications and Knowledge:** Key products will include a series of briefs on the API / decision tree integration process, case studies on bundled service models with private sector partners, and an impact brief from the upcoming MELIA study. Success stories will be shared via the CGIAR Results Dashboard and program updates.

## 8. Contact

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